Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Structures for Energy Absorption

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Tensegrity-inspired architectures combine rigid compression members with a network of soft tension elements, yielding lightweight assemblies with tunable nonlinear responses that are attractive for impact mitigation in aerospace landing structures and related protective applications. Multi-material fused-deposition modeling (FDM) of polylactic acid (PLA) struts and thermoplastic polyurethane (TPU) tension elements makes it possible to fabricate pre-assembled tensegrity-inspired geometries rapidly enough to support an experiment-driven design search. We report a closed-loop Bayesian optimization (BO) framework in which a fully Bayesian Gaussian-process surrogate is trained directly on instrumented drop-tower measurements and used to recommend the next batch of designs to print and test. A three-strut prism family is parameterized by five continuous geometry variables under a constant-mass fairness constraint, and each printed specimen is characterized by more than one hundred repeated drops. The campaign minimizes two impact objectives: the filtered peak-acceleration ratio across the specimen and a mass-weighted score for the rebound motion returned to the payload. We report the nine-specimen quasi-random seed round, the resulting surrogate diagnostics and Pareto set, and the first model-recommended batch, and we describe a metal-analog validation protocol that tests whether learned design rankings transfer across material systems.

Keywords: tensegrity, multi-material 3D printing, Bayesian optimization, energy absorption, design of experiments

1 Introduction

Tensegrity structures are assemblies of rigid compression members suspended within a continuous network of tension members; their defining property, that no two rigid bars touch, yields lightweight assemblies with remarkable strength-to-weight ratios and tunable nonlinear mechanical responses [1, 2]. These properties have made tensegrity attractive across scales, from metamaterial unit cells [3–5] to compliant impact-absorbing robots developed for planetary landing [6–10]. Recent work has shown that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact [11], opening a path toward additively manufactured architected absorbers. Throughout, we use *tensegrity-inspired* to denote printable unit cells that adopt the characteristic tensegrity motif (rigid compression members carried by a network of pretensioned soft tension elements) while relaxing the strict classical requirement that no two compression members touch and that the tension members behave as ideal inextensible cables; here the tension elements are extruded TPU and the struts may share printed junctions.

Multi-material fused-deposition modeling (FDM) that pairs a rigid polymer such as polylactic acid (PLA) with a soft, rate-dependent elastomer such as thermoplastic polyurethane (TPU) has been used to fabricate rigid–soft architected energy absorbers on a single platform: thick-panel origami metamaterials with PLA panels and TPU hinges [12] and ABS/TPU honeycombs with tunable stiff/flexible proportions [13]. The same rigid–soft co-printing capability makes it possible to rapidly fabricate diverse tensegrity-inspired geometries, which adopt a discrete-compression/continuous-tension topology rather than an origami or honeycomb one. Although extruded TPU does not behave as an idealized inextensible cable, its rate-dependent damping is a desirable property for impact-absorption applications and complements the elastic stiffness of the PLA compression members rather than competing with it.

Designing such architected absorbers is fundamentally a *design-of-experiments* problem: the configuration space spanned by strut geometry, tension-element cross-section, connectivity, and unit-cell tiling is large; each candidate must be physically printed and tested; and the relevant impact-performance metrics (the shock transmitted across the structure and the energy returned to the protected payload) are noisy and partially correlated. Bayesian optimization (BO) is well-matched to this regime, having driven breakthroughs from hyperparameter tuning of AlphaGo [14] to sustainable concrete [15], autonomous discovery of organic laser emitters [16], and architected materials [17–20]. The gap this paper closes is the combination: Pajunen et al. [11] printed tensegrity-inspired cells under a single fabrication condition with no optimization loop, and Mo et al. [17] closed a multifidelity BO loop entirely in simulation; to our knowledge no prior study closes a BO loop over co-printed multi-material tensegrity-inspired hardware using physical impact measurements as the sole ground truth.

We emphasize one framing point at the outset: the printed PLA/TPU T_3 prism studied here is a testbed for developing and validating the closed-loop workflow, not flight hardware. The planetary-lander setting motivates the objectives (limit the shock delivered to a payload, limit the energy returned to it) and the constant-mass fairness constraint, but the contribution is the loop itself, which is agnostic to the eventual application scale.

Contributions. This paper makes the following contributions:

(1) A parameterized family of multi-material 3D-printable tensegrity-inspired unit cells with PLA compression mem-

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August 22, 2026

bers and TPU tension elements that are anchored *inside* the ends of each PLA strut (the strut acting as a rigid cage in which the cables meeting at a given end join before exiting through discrete outlets), a joint geometry intended to improve cyclic interface durability, to be verified by pull-out and fatigue testing. Candidate designs are compared fairly through a constant-mass projection that rescales every geometry to a common material budget before printing (Section 3.1).

(2) A characterized drop-tower protocol and objective stack for printed impact absorbers: 101 repeated instrumented drops per specimen, standardized SAE J211 filtering [21], a velocity-change rig-health gate, and two per-specimen objectives (the filtered peak-acceleration ratio across the specimen and a mass-weighted rebound score), each characterized for repeatability before entering the optimizer (Section 3.3).

(3) An experiment-driven BO loop, built on BoTorch/Ax [22, 23] with a fully Bayesian sparse-prior surrogate (SAASBO) [24], that ingests the physical drop measurements, models per-print mass explicitly, and recommends the next batch of designs to fabricate; the seed round and first recommended batch are reported here (Sections 3.4 and 4).

(4) A physics-based simulation ladder (rigid-body through finite-element tiers) used for screening and hypothesis generation alongside the physical loop, with its predictive reach and blind spots quantified against the measured seed round (Section 3.5).

(5) A pre-registered metal-analog validation protocol (hollow aluminum struts with threaded stainless-steel cables) that tests whether the learned design ranking transfers across material systems, spanning predicted worst, mediocre, and best designs rather than only the optimum (Section 3.6).

The present study is deliberately scoped to single-cell, fixed-topology T_3-prism specimens under *axial* quasi-static and drop-weight loading. Off-axis and combined loading, multi-hit and cyclic durability, and landing-attitude variability are outside the scope of this paper and are identified as future work (Section 5); the novelty and generality claims below are correspondingly scoped to this restricted prototype design space rather than to tensegrity-inspired absorbers in general.

The remainder of the paper is organized as follows. Section 2 reviews tensegrity mechanics, multi-material 3D printing for energy absorption, and Bayesian optimization with an emphasis on multi-objective and noise-robust acquisition functions. Section 3 introduces the design parameterization, fabrication workflow, drop-tower protocol and objectives, BO loop, simulation screening ladder, and metal-analog validation plan. Section 4 reports the seed round and the round-1 surrogate, and Section 5 discusses the implications and limitations of the approach. Section 6 concludes.

2 Background

2.1 Tensegrity-Inspired Architectures for Energy Absorption. Tensegrity prisms and their planar extensions exhibit softening–stiffening response under compression [3–5], characteristics desirable for impact mitigation because the structure can absorb a large fraction of the input energy at a controlled, sub-injurious force plateau. Santos [25] recently characterized a tensegrity energy-dissipation metamaterial reporting equivalent viscous damping on the order of ~23% with negligible residual strain per impact, underscoring the suitability of tensegrity-inspired architectures for repeated-impact use cases. The space program literature documents large-scale tensegrity-inspired absorbers for planetary landing [6–10] and a long history of crushable energy-absorbing landing systems [26–28]. Pajunen et al. [11] demonstrated that truncated-tetrahedral tensegrity-inspired unit cells can be directly 3D-printed and exhibit load-limiting plateaus, motivating the present focus on additively manufactured TPU/PLA composites. More recent work on dynamically loaded additively-manufactured energy-absorbing lattices [29], together with Intrigila et al.'s bistable tensegrity-like unit cell for lattice metamaterials [30] as the closest published analog, confirms that this architecture class also responds favorably under impact and quasi-static loading, although neither study employed multi-material FDM or closed-loop design optimization. Table 1 situates the present work against the three most closely related prior efforts, making explicit that the combination of a multi-material FDM tensegrity-inspired architecture with an *experiment*-driven optimization loop is, to our knowledge, not addressed by any single prior study.

2.2 Multi-Material 3D Printing of PLA/TPU Composites. Multi-material rigid–soft FDM enables stiff/compliant combinations on a single platform with tunable energy absorption, demonstrated for thick-panel origami with PLA (or ABS/CFRP) panels and TPU hinges [12] and for ABS–TPU honeycombs [13]. Neither study is itself a tensegrity, but both establish the rigid–soft co-printing route we adopt. Ye et al. [12] introduced a *wrapping-based* strategy in which rigid cores are encapsulated by continuous soft skins, reported to reduce interface delamination under cyclic loading, something we take inspiration from in our designs. Beyond these rigid–soft demonstrations, a growing body of work confirms that genuinely tensegrity-inspired architectures can be additively manufactured and exhibit programmable mechanical response. Bauer et al. [31] showed that 3D-printed tensegrity metamaterials delocalize deformation to resist catastrophic failure, while Pajunen et al. [32] tuned acoustic bandgaps in 3D-printed tensegrity-inspired lattices through prestrain, and Sabouni-Zawadzka et al. [33] experimentally characterized 3D-printed tensegrity-inspired metamaterials built on the simplex module. More directly relevant to impact, Almeida et al. [34] and Davami et al. [35] characterized the high-strain-rate and dynamic response of additively manufactured tensegrity-inspired structures, and Wang et al. [36] reported an integrated fabrication-and-validation workflow for tensegrity-inspired rigid–flexible metamaterials. Collectively these studies establish the additive-manufacturability, tunability, and impact relevance of the architecture class adopted here, none combining multi-material FDM with an experiment-driven optimization loop. Recent multi-material PLA/TPU sandwich and layered composites [37, 38] report quantitative bounds on stiffness, strength, and energy absorption for the same PLA–TPU pair used here, while FDM PLA fatigue data [39, 40] inform conservative design bounds for the rigid struts. Architected multi-material lattices with co-printed compliant skins have been explicitly designed for high specific energy absorption [41], with FDM-specific multimaterial interface adhesion (notably TPU bonded to rigid co-printed substrates) characterized in [42]; recent surveys catalogue the rapidly expanding additively manufactured polymeric energy-absorption literature [43, 44].

2.3 Bayesian Optimization for Architected-Material Design. Bayesian optimization is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian-process surrogate [45, 46]. Modern implementations such as BoTorch [22] and the Ax platform built atop it [23] support parallel batched evaluations and a wide variety of acquisition functions; recent algorithmic advances include log-EI for numerical robustness [47], noisy expected hypervolume improvement for multi-objective settings [48], input-noise-robust acquisitions [49], and evolution-guided constrained multi-objective BO for self-driving labs [50]. Mixed/categorical search spaces, needed for the planned connectivity-topology extension rather than the continuous T_3-prism prototype studied here, have dedicated treatments in BOCS [51] and the one-hot-with-rounded-kernel construction of [52]. BO has been applied to mechanical metamaterial and structural design [17–19] and to additive manufacturing [20]. Mo et al. [17] in particular established a multifidelity BO framework for architected materials by fusing low-fidelity simulations with high-fidelity ground truth via nonlinear information-fusion priors [53];

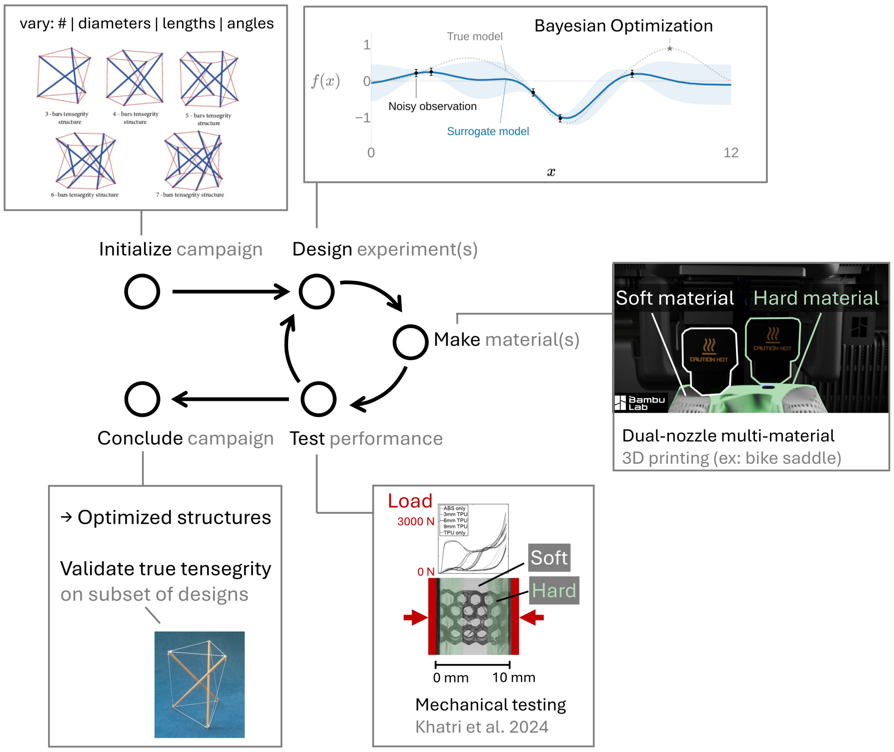

Fig. 1 Closed-loop, experiment-driven design framework. Parameterized PLA/TPU tensegrity-inspired unit cells are printed, characterized by repeated instrumented drop-weight impacts, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs to fabricate.

the present work shifts the emphasis from simulation surrogates to direct experimental data, appropriate for materials whose constitutive response is difficult to calibrate from first principles.

3 Materials and Methods

3.1 Design Parameterization. We parameterize a family of tensegrity-inspired unit cells via four groups of design variables. For the working T_3-prism prototype that defines the initial BO search space (Section 3.4), only the first two groups are active and all of their variables are *continuous*; the connectivity-topology and tiling groups are held fixed for this prototype and are listed here as planned extensions of the design family:

- **PLA compression members.** Strut diameter d_s and length ℓ_s, with the slenderness ratio ℓ_s/d_s bounded away from buckling-dominated regimes.
- **TPU tension elements.** Cross-sectional area A_t (or equivalent diameter d_t for circular cross-sections).
- **Connectivity topology (future extension).** Selected from a discrete set of candidate unit cells (e.g., truncated-tetrahedral, prism-based, and octahedral variants); when activated, the design vector would encode this as a categorical choice. It is held fixed at the T_3 prism for the present campaign.
- **Unit-cell tiling (future extension).** Number of cells $N_x \times N_y \times N_z$ and the orientation of each cell relative to the impact direction; held fixed at a single cell here.

Working prototype. The first instance of this family used to validate the fabrication and test pipeline is a three-strut (T_3) tensegrity prism scaled to a ~50 mm bounding box. As implemented in the BO harness, this working prototype exposes five continuous design variables (base radius R, cell height H, twist angle θ, a single strut diameter d_s, and a single cable (tendon) diameter d_t) over the ranges listed in Table 2; the cable diameter d_t is treated as a *continuous* variable on $[3.0, 5.5]$ mm rather than a discrete set. Under the prism's D_3 symmetry all struts share one diameter and all cables share one diameter, so the present parameterization uses a single strut-diameter axis and a single cable-diameter axis (two member-diameter axes), leaving room to relax to per-orbit (saddle, top, bottom) or fully per-member diameters once the individual orbits have been characterized. The joint diameter d_j is held fixed at 7.0 mm rather than optimized: it sets the printed strut-end cage that houses the internal cable anchors (Section 3.2), so it is constrained primarily by the TPU-tendon outlet count and the dual-nozzle clearance rather than by energy-absorption performance, and fixing it keeps the anchor geometry constant across specimens so that the five optimized variables are not confounded by joint-strength changes; relaxing d_j is deferred to a follow-on joint-design study. Figure 2 shows the parametric CAD model alongside an as-printed prototype of this working T_3 prism.

Constant-mass projection and printability screens. Because cell mass varies severalfold across the raw design box, an unconstrained search would partly reward designs for carrying more material rather than for their architecture. Every candidate is therefore projected onto a constant material budget before printing: the design is uniformly rescaled (all dimensions except the twist angle) until its solid CAD mass $\rho_{\mathrm{PLA}}V_{\mathrm{PLA}} + \rho_{\mathrm{TPU}}V_{\mathrm{TPU}}$ equals $m^\star = 30.95$ g, the solid mass of the reference prism, converged to within 0.15 g on rendered geometry. Two printability screens are evaluated on the projected geometry and recorded as flags rather than hard rejections: a cylindrical build envelope $\pi R^2 H \leq 250\,\mathrm{cm}^3$ and the 3.0 mm TPU self-bridging floor on the as-projected cable diameter (the projection can scale a nominally feasible cable below the floor; the two seed designs where this occurred both exhibited

Table 1 Positioning of this work against the most closely related prior efforts. The present study is distinguished by combining a multi-material (PLA/TPU) FDM tensegrity-inspired architecture with a closed-loop optimizer trained directly on physical measurements rather than simulations or a single fabrication condition.

Study	Architecture	Fabrication	Optimization	Ground truth
Pajunen et al. 2019 [11]	Truncated-tetrahedral tensegrity-inspired	Single-material 3D print	None (single condition)	Experiment (drop impact)
Intrigila et al. 2022 [30]	Bistable tensegrity-like lattice unit cell	Single-material 3D print	None	Experiment (quasi-static/impact)
Mo et al. 2023 [17]	Architected lattices	n/a (simulation)	Multifidelity BO	Simulation (+ sparse high-fid.)
This work	Tensegrity-inspired T_3 prism (PLA/TPU)	Multi-material FDM (PLA + TPU)	Experiment-driven BO (SAASBO/qNEHVI)	Physical experiment

tendon stringing defects, consistent with the screen). The choice of a *solid-mass* budget has a documented limitation discussed in Section 5: PLA struts print at partial infill while thin TPU cables print near solid, so as-printed masses still ranged from 18.50 g to 22.29 g across the seed batch; a calibrated printed-mass model that closes this gap is described in the Supplementary Information.

Table 2 Design variables for the T_3-prism campaign, as implemented in the BO harness. The seed round is an $n = 9$ Sobol draw (fixed seed) over this box, with each design then projected onto the constant solid-mass budget $m^\star = 30.95$ g. The joint diameter is held fixed; all specimens are printed in the vertical orientation on a Bambu Lab H2D, one specimen per named build plate.

Design variable	Symbol	Units	Range
Base radius	R	mm	25–40
Cell height	H	mm	60–110
Twist angle	θ	deg	40–80
Strut (PLA) diameter	d_s	mm	6.0–12.0
Cable (TPU) diameter	d_t	mm	3.0–5.5
Joint diameter (fixed)	d_j	mm	7.0

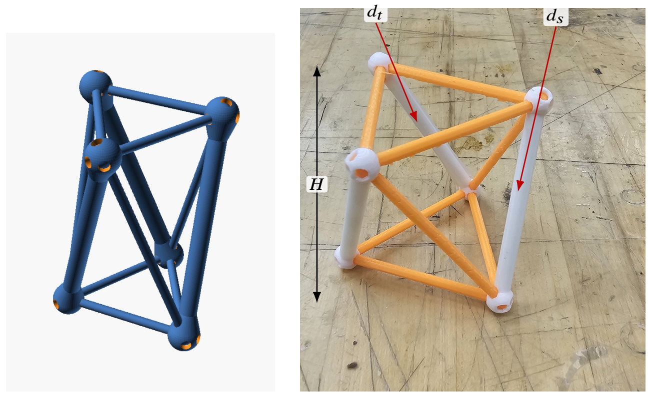

Fig. 2 Working T_3-prism prototype. *(a)* Parametric OpenSCAD model whose base radius R, cell height H, twist angle θ, strut diameter d_s, and cable diameter d_t are the design variables of Table 2. *(b)* A representative dual-material print from the PR #35 print history (rigid PLA struts, here in white; tendon members in orange), with callouts marking the strut diameter d_s, a tendon (cable) diameter d_t, and the cell height H; the inter-plate twist θ is the relative rotation of the top and bottom node triangles, and the base radius R sets their circumradius.

As-built resolution of the continuous variables. Although d_s and d_t are treated as continuous, the as-built cross-sections are discretized by the nozzle diameter, the slicer line width, and support interaction, so the effective search granularity is coarser than the nominal continuous box. The realized member diameter therefore changes in steps set by the line-width quantization rather than continuously, and the smallest CAD change that yields a reliably distinguishable as-built member is bounded below by these slicer limits; the lower TPU-tendon bound (3.0 mm) is itself a workflow-specific limit tied to the TPU self-bridging and auto-support detection floor on this H2D platform rather than a fundamental geometric one. The as-built quantity measured for every article is its mass, weighed on a 0.01 g scale; dimensional metrology of the printed members has not yet been performed, so the geometry attached to the surrogate is the commanded post-projection geometry, and the print-to-print spread (mass standard deviation 0.457 g across the one design printed in triplicate) is carried into the observation noise model (Section 3.4).

3.2 Multi-Material Fabrication. Specimens are fabricated on a multi-material FDM printer using PLA for the rigid struts and TPU for the soft tension elements. Figure 3 summarizes the end-to-end fabrication and characterization workflow, from parametric CAD through multi-material slicing, dual-extrusion printing, post-processing, and mechanical testing. Rather than wrapping the struts in a continuous TPU skin, the current design anchors the TPU tension elements *inside* the ends of each PLA strut: the cables converging at a given strut end are joined within the strut, which acts as a rigid cage housing the junction, and the individual tendons then exit through discrete outlets. This keeps the soft–rigid interface in internal, compression-dominated pockets rather than at an exposed overmolded surface. This construction effectively inverts the material roles of Ye et al.'s soft-skin-over-rigid-core wrapping [12]: here a soft TPU core is captured within a rigid PLA shell, so we cite that work only as a multi-material co-printing precedent rather than as validation of the present junction.

Print platform and joint geometry. The working platform is a Bambu Lab H2D printer, which permits a single rigid–soft build without filament swaps. Strut endpoints are tied to cables through parametric joint geometries developed and ranked through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants); the working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies, with a captive TPU core routed inside the PLA shell to keep the soft tendon protected from layer-line failure at the strut-to-cable transition. The full five-design comparison and the selected junction are detailed in the Supplementary Information.

Supports for soft members. Because near-vertical TPU tendons are otherwise unsupported during printing, automatic support generation is disabled and supports are painted manually in the slicer: PLA edge supports run along each tendon, with the support-toobject clearance tightened from 0.35 mm to 0.2 mm during tuning on the reference prism. Two support-removal artifacts recur in the seed batch (surface stringing and, on two articles, TPU strands pulled off bottom tendons during support removal) and are logged per article in the print key (Supplementary Information). An automated alternative (a lowered support-threshold angle plus mesh-ray-cast narrowing pillars) was developed in parallel and is documented in the Supplementary Information.

Moisture and nozzle management. TPU is hygroscopic, so the soft filament is fed from a heated dry box held near 8 % relative humidity, logged per print. Midway through the seed batch the TPU extruder developed a progressive flow taper (a partial clog, compounded by a stray PLA fragment found inside the extruder); it was resolved by a cold pull and by moving the TPU to a dedicated high-flow 0.6 mm nozzle, after which print quality was restored. The affected and post-fix articles are identified in the print key so that any nozzle-related covariate can be checked against the measured objectives.

Table 3 Multi-material print parameters on the Bambu Lab H2D. Cells marked *TBD* are pending confirmation; the reporting structure follows the methods section of Ye et al. [12].

Parameter	PLA (struts)	TPU 85A (tendons)
Nozzle	0.4 mm	0.6 mm high-flow
Nozzle temperature (°C)	*TBD*	*TBD*
Bed temperature (°C)	*TBD*	
Layer height (mm)	*TBD*	
Infill	15% grid, 2 walls	near-solid
Filament drying	n/a	dry box, ~8% RH
Build-plate type	Textured PEI	
Build orientation	Vertical	
Supports	Manual painted PLA edge supports	

3.3 Experimental Characterization.

Drop-tower fixture. The primary characterization is repeated drop-weight impact on a Lansmont Model 23 shock test system. The printed specimen rides upright on the carriage plate, which is raised 60 in (1.524 m) along guide rails and released; the plate lands on a single 0.5 in Shore 60 OO polyurethane cushioning sheet, which shapes the landing into a single sharp base pulse of roughly 1.6 ms duration. This impact-surface configuration was selected by a blinded crossover experiment over three candidate mat arrangements, and unlike a felt stack it does not measurably compact over hundred-drop sessions. Rig health is monitored continuously through the impact velocity change Δv recovered from the base accelerometer: over an uninterrupted 100-drop characterization session the Δv coefficient of variation was 0.54% with no drift, and every campaign session is gated on Δv remaining at the healthy tower baseline. One drop takes roughly 42 s, so a full 101-drop specimen characterization completes in about ninety minutes.

Instrumentation and signal processing. A single-axis piezoelectric accelerometer on the carriage plate records the base input and triggers the acquisition; a triaxial accelerometer (Dytran 3133A4) seated in a printed pocket at the specimen's top vertex records the transmitted response. The two base-capable channels were cross-calibrated in place over fifteen co-located drops (ratio 0.953 ± 0.001 at matched filtering), and the transducer settings are archived with the data. Signals are captured at 1.25 MHz over a 100 ms record with a 2 ms pre-trigger. All channels are low-pass filtered with the SAE J211/1 channel-frequency-class (CFC) filters [21], implemented from the standard's published coefficients with the two-pass corner frequency handled explicitly (a common implementation shortcut narrows the passband by roughly 20%, which we identified and corrected before the campaign; all reported numbers use the corrected filter). CFC-180 is the reporting class for peak metrics, with CFC-1000 retained as a higher-bandwidth diagnostic.

Per-specimen objectives. Each 101-drop session (first two drops discarded as warm-up; one article's session recorded 103 captures) yields two objectives, each reported as a stabilized mean with the per-drop scatter quoted as one standard deviation; the standard error of the mean is what enters the surrogate's noise model (Section 3.4):

$$t_{180} = \frac{\max_t \|\mathbf{a}_{\mathrm{top}}(t)\|_{\mathrm{CFC180}}}{\max_t |a_{\mathrm{base}}(t)|_{\mathrm{CFC180}}}, \qquad E_{\mathrm{reb}} = e_{\mathrm{reb}}\, m\, g\, h, \qquad e_{\mathrm{reb}} = \frac{g\, t_{\mathrm{second}}}{2\,\Delta v}, \tag{1}$$

where t_{180} is the CFC-180 *filtered peak-acceleration ratio*: the peak magnitude of the triaxial top-vertex response over the peak base input within the impact window. We deliberately do not call this quantity a transmissibility: the two peaks need not be simultaneous and a single transient drop cannot produce the averaged frequency-response function that the standards-defined transmissibility requires, so t_{180} is a filter-class-specific ranking metric rather than a standards quantity, and SAE J211/1 is cited here only for the impact-channel filtering. Values above unity indicate the structure amplifies the peak shock reaching its top vertex; values below unity indicate attenuation. E_{reb} is a *provisional mass-weighted rebound score* with units of energy (millijoules per drop): the dimensionless rebound fraction e_{reb} is recovered from the ballistic timing of the top vertex's post-impact hop [54] (t_{second} is the time to the second contact event and Δv the measured base velocity change), and it is scaled by each article's own weighed mass m and the drop height h. Two interpretive caveats are deliberate. The ballistic reading of the second contact event has not yet been validated against synchronized video, and Δv is the base plate's velocity change rather than a demonstrated incident relative velocity, so e_{reb} is a rebound-timing score rather than a coefficient of restitution; and because a strict restitution reading would scale returned kinetic energy as the *square* of a velocity ratio, E_{reb} is a mass-aware ranking score that penalizes designs (and grams) that throw more motion back at the payload, not a measured rebound energy. The score is used as the round-1 second objective and is slated to become a constraint once validated (Section 5). Using the mass-weighted form rather than the bare fraction keeps the objective mass-aware: equal fractions on articles of different printed mass are not equivalent outcomes for a payload. Ringdown descriptors (the fitted natural frequency $f_n \approx 300$ Hz to 470 Hz and, where the exponential fit passes an $r^2 \geq 0.85$ gate, a damping ratio ζ) are recorded per drop as print-health and prestress diagnostics [55] but are not campaign objectives. Within-specimen repeatability of t_{180} is 0.2 to 0.5% (CV), and the one article re-tested in a separate session reproduced its mean to 0.13%. Figure 4 illustrates the mechanism-oriented presentation we adopt for the processed traces, linking features of the measured signal to the structural events that produce them.

Planned quasi-static extension. The drop test loads the specimens in their elastic regime (the crush lasts 1 to 3 ms and the article recovers), so crush-energy metrics such as specific energy absorption (SEA) and compaction efficiency are not recoverable from these captures by any reprocessing. A quasi-static compression campaign on a benchtop load frame (BYU CB 123 Instron, with a low-range load cell appropriate to the expected cell stiffness) is planned to supply the complementary crush metrics

$$\mathrm{SEA} = \frac{1}{m}\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta, \qquad \eta_c = \frac{\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta}{F_{\max}\,\delta_d}, \tag{2}$$

with the densification displacement δ_d taken at the maximum of the energy-absorption efficiency curve and $F_{\max}$ the peak force within $[0, \delta_d]$, following the canonical cellular-solid treatments [56–59]

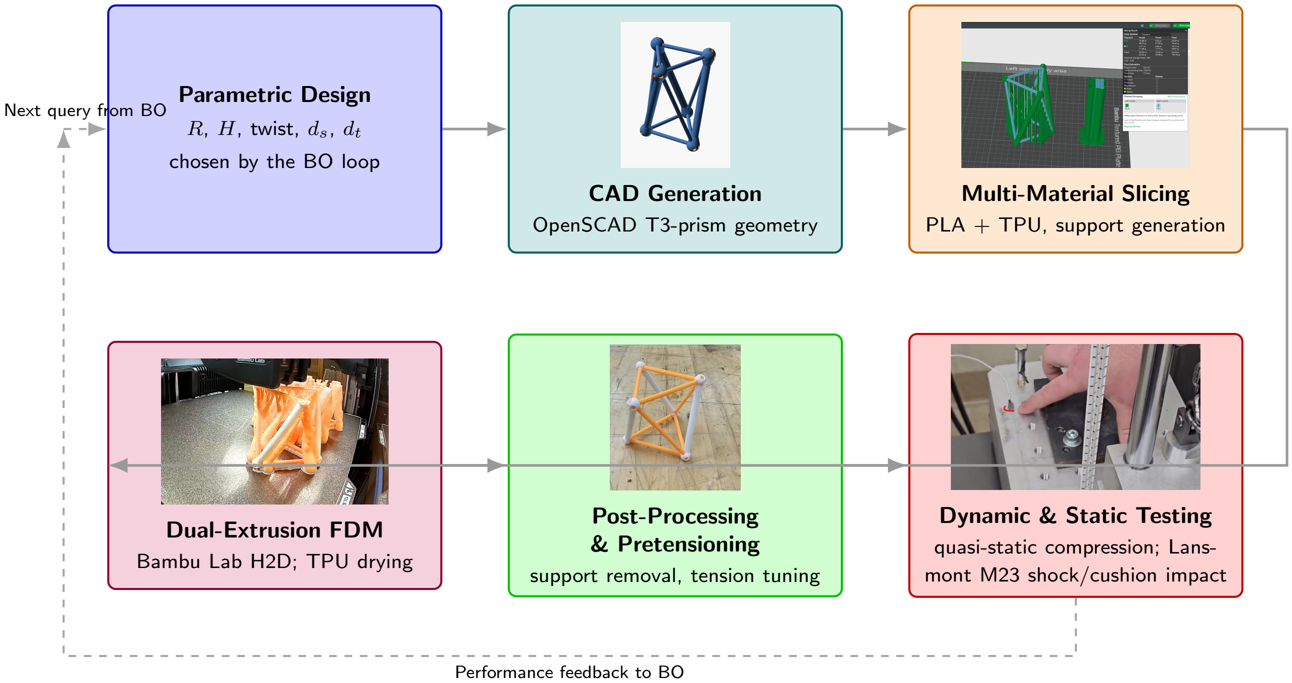

Fig. 3 Fabrication and characterization workflow for the multi-material tensegrity-inspired unit cells: design parameters drive parametric CAD, slicing with manually generated TPU supports, a single multi-material print on the Bambu Lab H2D, post-processing and inspection, and mechanical testing. Node photographs are representative artifacts drawn from the project's build history: the OpenSCAD T_3-prism render (CAD), the dual-nozzle Bambu Studio slice with PLA on the left extruder and TPU on the right (slicing), an in-progress H2D print of the prism batch (FDM), an as-printed specimen (post-processing), and the bungee-assisted drop-tower base plate with accelerometer (testing).

and mirroring the quasi-static protocol of Pajunen et al. [11]. These metrics are reported as a complementary characterization of selected designs rather than as objectives of the drop-tower campaign. Slip resistance and traction at the specimen-to-floor interface, governed by the tip geometry rather than the internal architecture, follow ISO 11334-4 [60] for walking-aid tips and are out of scope for the present axial-impact study.

Complementary data streams. High-speed video (959 fps, three clips per specimen) supplements the accelerometer data. The camera cannot resolve the millisecond-scale impact pulse (one to two frames), so the division of labor is explicit: the accelerometer channels produce the objectives, and video documents the post-impact rebound and recovery. Synchronized laser-Doppler vibrometry is reserved for follow-up characterization of selected designs.

3.4 Experiment-Driven Bayesian Optimization Loop. The optimization loop maintains a Gaussian-process (GP) surrogate over the design space defined in Section 3.1. After each round of physical experiments, the surrogate is refit and an acquisition function is maximized to recommend the next batch of designs to fabricate. The campaign is posed as a two-objective minimization: simultaneously minimize the filtered peak-acceleration ratio t_{180} and the per-drop rebound score E_{reb} of Eq. (1). Both objectives serve the same payload: t_{180} bounds the peak shock it feels during the impact, and E_{reb} penalizes the motion thrown back into it afterward. The seed data show an apparent trade-off driven substantially by the single strongest attenuator; at seven mapped articles it is not statistically resolved (Spearman $\rho = -0.39$, $p = 0.38$; Section 5), which is why the rebound axis is slated to become a constraint rather than an objective once validated. Implementation uses the Ax platform [23] over BoTorch [22], with noisy expected hypervolume improvement (qNEHVI) [48] as the multi-objective acquisition. The working T_3-prism search space is fully continuous (the five variables of Table 2), so no categorical encoding is required; the mixed-search-space constructions of [52] and the BOCS combinatorial baseline [51] remain the route for the future connectivity-topology categorical rather than part of the current loop. The input-noise-robust acquisition of Daulton et al. [49] and the evolution-guided constrained variant of Low et al. [50] are held in reserve as contingencies.

Surrogate and noise model. The surrogate is the fully Bayesian sparse axis-aligned subspace model (SAASBO) [24]: a Matérn-5/2 kernel with automatic relevance determination whose length scales carry strong sparsity-inducing priors and are marginalized by Markov-chain Monte Carlo rather than point-estimated. We retain SAASBO even though the dimensionality is modest because marginalizing the hyperparameters tends to improve uncertainty quantification at the very small sample sizes of physical campaigns; the fitted model is audited each round with leave-one-out cross-validation and length-scale-based sensitivity diagnostics against a standard single-task GP with the same kernel (Section 4). Inputs are normalized to the unit cube and outputs standardized. Observation noise is supplied per observation as a fixed variance: for t_{180} the squared standard error of the stabilized mean, $\sigma_t^2 = s^2_{\mathrm{drop}}/n_{\mathrm{drops}}$, and for the rebound score the drop-scatter term plus the propagated print-to-print mass scatter, $\sigma_E^2 = \sigma^2_{E,\mathrm{drops}} + (\partial E_{\mathrm{reb}}/\partial m)^2 \sigma_m^2$ with $\sigma_m = 0.457$ g measured from the one design printed in triplicate. The estimand this noise model implies is the response of the tested article, not of a freshly printed article at the same design: an independent adversarial review (Section 5) argues the decision-relevant target is the latter, for which an article-level noise floor near 0.7% CV (14 to 44 times the per-drop standard error) should replace the per-drop terms; the round-3 refit adopts that correction. SAASBO is the prespecified round-1 surrogate; we make no claim that it outperforms a stan-

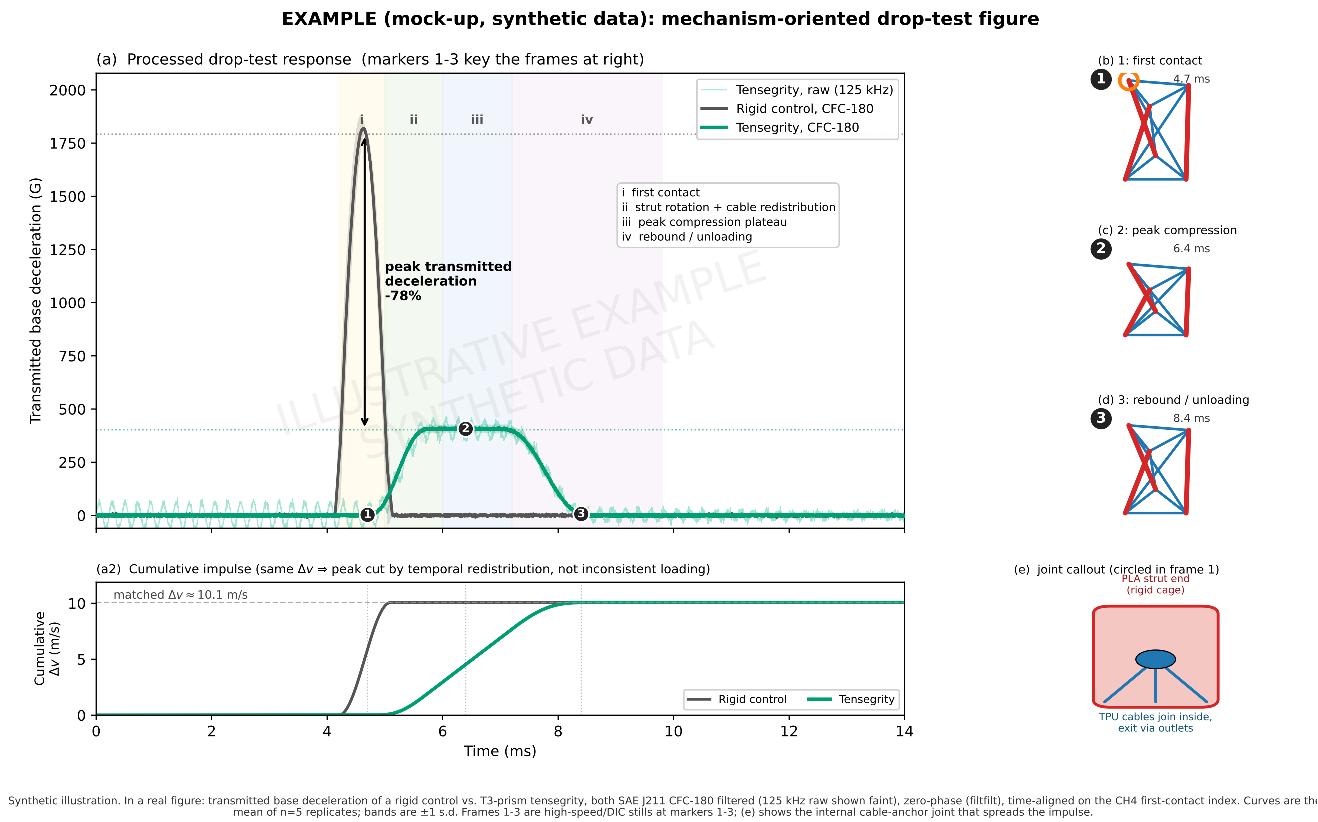

Fig. 4 ***Illustrative example (synthetic data).*** **Template for the mechanism-oriented drop-test figures planned for the Results: (a) processed deceleration response of a rigid control vs. a tensegrity specimen (SAE J211 CFC-180 filtered [21]), with event-based phase shading and numbered markers keyed to the schematic frames (b)–(d); (a2) cumulative-impulse subpanel showing the two traces transfer the same mass-normalized Δv, so the lower tensegrity peak reflects temporal redistribution rather than inconsistent loading; (e) the internal-anchor joint load path. Curves are synthetic and impulse-consistent (anchored to a peak from an early commissioning-era capture, which is why the amplitude differs from the campaign values in Table 5); to be regenerated from the campaign captures. Generated by `scripts/figures/mechanistic_data_figure_example.py`.**

dard single-task GP at this dimensionality and sample size, and the budget-matched calibration comparison remains a pending audit (Section 4).

Mass treatment. Because the constant *solid*-mass projection does not hold printed grams constant (Section 3.1), printed mass correlates strongly with the shape variables in the seed round ($r = 0.83$ against t_{180} across the mapped articles). The loop therefore makes mass explicit rather than pretending it away: the rebound objective is expressed in absolute millijoules using each article's weighed mass, so added grams are re-penalized through the energy they deliver back to the payload, while t_{180} deliberately stays a ratio because payload peak acceleration is a damage criterion that does not scale with specimen mass. The objective surrogates take the five design coordinates of Table 2 as their inputs; printed mass is attached as a separately modeled *tracking metric*, so the model learns as-printed mass as a function of the design coordinates and every recommendation reports a predicted printed mass (the diagnostic refit behind Fig. 8 additionally includes printed mass as a sixth input, precisely to probe this confound). A calibrated printed-mass model that would let future rounds hold *printed* grams constant is described in the Supplementary Information.

Budget and seeding. The seed round is an $n = 9$ Sobol draw with a recorded seed over the five-variable box, printed together with the fixed reference prism (S0). Optimization proceeds in sequential batches of $q = 9$ prints, sized to one build-plate generation cycle; the first model-recommended batch (Ax trials 10 to 18) is reported in Section 4. All random seeds, the full experiment state, and the recommendation tables are recorded per round; full computational reproducibility (the exact processing pipeline and model settings) is carried by the campaign code, whose archival release is a declared submission gate rather than a solved item. Escalation to trust-region BO (TuRBO) [61] is planned only if per-member-diameter extensions push the design space well beyond the present regime; a single-objective log-Expected-Improvement run [47] on t_{180} is retained as a baseline for quantifying the benefit of the multi-objective formulation. The campaign code and recommended batches will be archived at a Zenodo-minted DOI mirroring the GitHub repository.

3.5 Simulation Screening Ladder. Alongside the physical loop we maintain a three-tier simulation ladder used for screening and hypothesis generation, never as ground truth. Tier C (rigid struts with spring tendons in a rigid-body dynamics engine, roughly 0.15 s per design) ranks large design sets; Tier B (deformable struts with explicit tendon cross-sections, seconds to tens of seconds per design) restores the tendon-stiffness sensitivity that Tier C is blind to; and Tier A (hyperelastic finite-element contact simulation, about a minute per design after meshing) captures strut buckling and self-contact for a handful of designs. The ladder applies the same SAE J211 CFC-180 filtering as the physical pipeline so simulated and measured records are processed identically. Its honest scorecard against the measured seed round is mixed and is reported as such: an intensive volumetric energy-absorption score computed at Tier C is a strong *cross-metric* predictor of the measured t_{180}

ranking (Spearman $\rho = -0.93$ across the seven mapped articles), yet the ladder's own simulated t_{180} correlates only weakly with the measurement ($\rho \approx 0.5$), rigid-strut models cannot reproduce the observed amplification ($t_{180} > 1$ on five of eight measured articles), and simulated restitution is design-invariant where the measured rebound is not. We therefore use the ladder to prioritize candidates and probe mechanisms (for example, the mass-confound diagnosis in Section 5) while keeping every surrogate observation physical. Full engine benchmarks and the simulation-side BO study, in which the same acquisition pipeline run on the Tier-C evaluator reaches 97% of an exhaustively computed reference hypervolume while random and space-filling baselines reach 61 to 66%, are provided in the Supplementary Information.

3.6 Metal-Analog Validation. To test whether the rankings learned on the printed PLA/TPU specimens generalize beyond the as-printed polymer system, a final validation campaign reproduces a subset of the T_3-prism designs as *metal analogs* built from hollow aluminum rods (compression members) and stainless-steel threaded cables (tension members), using the same R, H, θ, and member-diameter parameterization. Rather than fabricating only the top-ranked geometries, we deliberately span the predicted performance range: a few designs the surrogate predicts to be *worst*-performing, a few *mediocre*, and a few *best*-performing are each built in both material systems. Comparing the measured ordering of the PLA/TPU specimens against their hollow-aluminum/stainless-steel counterparts tests whether both systems preserve the same rank ordering of energy-absorption performance, which would indicate that the optimization transfers across material systems rather than overfitting to the printed polymers. We treat this as a *stringent, exploratory* transfer test rather than a presumed one-to-one validation: the deformation modes, joint compliance, friction, pretension behavior, and strain-rate sensitivity all differ between the PLA/TPU and aluminum/steel systems, so agreement is not expected a priori. The rank-preservation criterion is fixed in advance as the Spearman rank-correlation coefficient ρ_s between the two material systems' orderings of the primary campaign metric (t_{180} under the identical drop protocol), with the rebound energy E_{reb} and, once the quasi-static extension runs, SEA reported as secondary rankings; ρ_s near unity would support cross-system transfer of the learned design ranking. Table 4 reports the planned rank-ordering comparison and Fig. 5 shows the assembled metal-analog and printed-polymer specimens for each performance tier.

Table 4 ***Placeholder (data pending).*** **Planned rank-ordering validation: representative T_3-prism designs predicted to be worst-, mid-, and best-performing, each fabricated as a printed PLA/TPU specimen and as a hollow-aluminum-rod / stainless-steel-cable metal analog. Entries record the measured performance metric (t_{180} under the identical drop protocol, with rebound energy secondary) and the resulting rank within each material system; agreement of the two rankings indicates the optimization transfers across material systems.**

Predicted tier	Design ID	PLA/TPU (rank)	Al/SS (rank)
Worst	*D–w1*	–	–
Worst	*D–w2*	–	–
Mediocre	*D–m1*	–	–
Mediocre	*D–m2*	–	–
Best	*D–b1*	–	–
Best	*D–b2*	–	–

Figure placeholder: Assembled validation specimens for each predicted performance tier (worst / mediocre / best), shown as printed PLA/TPU builds alongside their hollow-aluminum-rod and stainless-steel-threaded-cable metal analogs, with callouts marking the aluminum compression members, the threaded steel tension cables, and the strut-end cable anchors. Photographs to be added once the specimens are fabricated.

Fig. 5 ***Placeholder.*** **Assembled validation specimens for each predicted performance tier (worst / mediocre / best), shown as printed PLA/TPU builds alongside their hollow-aluminum-rod and stainless-steel-threaded-cable metal analogs, with callouts marking the aluminum compression members, the threaded steel tension cables, and the strut-end cable anchors. Photographs to be added once the specimens are fabricated.**

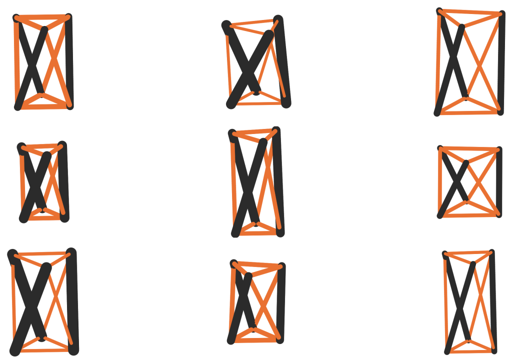

Fig. 6 The nine seed geometries rendered to a common millimeter scale (rigid PLA struts in black, TPU tendons in orange). Every rendering corresponds to a physically printed article in the seed campaign.

4 Results

4.1 Seed-Round Measurements. All nine Sobol specimens and the reference prism were printed and declared acceptable to test; eight articles (six mapped Sobol designs, the reference prism, and one article pending a design mapping) have completed 101-drop characterization sessions to date, totaling over 800 clean captures within a project-wide history of roughly 3,660 instrumented drops. Table 5 reports the stabilized objectives. Three findings frame the round. First, the measured t_{180} spans 0.893 to 1.062: five of eight articles *amplify* the peak shock reaching the top vertex (means above unity), two sit marginally below unity (0.997 and 0.980), and a single design (Sobol spec 01, the thick-strut, thin-cable, high-twist corner of the batch) attenuates it by 11%. The design-driven range of 0.169 in t_{180} (19% relative to the minimum) dwarfs the 0.2 to 0.5% within-specimen repeatability. Second, the objectives show an apparent trade-off in this sample: the best attenuator carries the largest rebound score (13.9 mJ per drop), so the observed front is nondegenerate, although the trade-off is driven substantially by that single article and is not statistically resolved at this sample size (Section 5). Third, printed mass correlates strongly with t_{180} ($r = 0.83$, $p = 0.02$, $n = 7$; an observational association confounded with the design coordinates, not a causal mass effect), which motivated the mass-aware objective formulation of Section 3.4. Figure 6 renders the seed geometries to a common scale, showing how widely the constant-mass box spans aspect ratio, member sizing, and twist.

4.2 Round-1 Surrogate, Pareto Set, and Recommended Batch. Figure 7 shows the measured objective space, the nondominated set, and the model-recommended round-2 batch at its posterior-mean predictions. The observed Pareto set comprises three articles (6lhxfy, 6nheas, and the reference prism bpx68c), and the model-predicted Pareto set selects the same three. The nine recommended designs (Ax trials 10 to 18) concentrate in the low-t_{180}

Table 5 Seed-round drop-tower results: 101 drops per article at 60 in onto the standardized polyurethane surface, CFC-180 filtering, first two drops discarded. Means ± one standard deviation over the stabilized drops. E_{reb} uses each article's weighed mass per Eq. (1). The ringdown frequency f_n is a print-health descriptor, not an objective. Article *amdjwm* lacks a confirmed design mapping and is excluded from the surrogate fit; spec 03 was tested but its session was not yet uploaded at the time of writing; specs 06 and 07 are printed and pending test.

Design	Article	Mass (g)	t_{180}	E_{reb} (mJ)	f_n (Hz)
Spec 01	6lhxfy	18.50	0.8931 ± 0.0042	13.9	368
Spec 05	6nheas	21.73	0.9970 ± 0.0032	13.1	322
(unmapped)	amdjwm	–	0.9805 ± 0.0025	–	455
S0 ref.	bpx68c	20.23	1.0111 ± 0.0024	6.2	468
Spec 00	9hhbkp	21.62	1.0183 ± 0.0017	7.0	364
Spec 04	nvxsrv	20.66	1.0275 ± 0.0044	8.2	294
Spec 02	autv5r	22.04	1.0404 ± 0.0035	8.8	386
Spec 08	bag26v	21.42	1.0616 ± 0.0051	7.7	–

region near and beyond the current best attenuator; they have been sliced and sent to the printer, and their measured outcomes will populate the predicted-versus-measured comparison in the next revision. The printability screens of Section 3.1 are diagnostic flags rather than acquisition constraints, and two recommended designs violate one screen each after projection; both were retained deliberately, in part to test whether the conservative screens predict actual fabrication failure (the two seed designs below the cable-bridge floor were exactly the two with tendon stringing defects).

Surrogate audit. Figure 8 reports the SAASBO feature importances (normalized inverse length scales, median over the Monte Carlo draws with interquartile whiskers, over the five design variables plus printed mass) and Fig. 9 the leave-one-out cross-validation. The honest reading at this sample size is that the sparsity prior has *not* identified a dominant subspace: no input separates decisively from the equal-share line (one sixth), with cell height leading for t_{180} and base radius for the rebound energy, and most interquartile bands straddling the line. Cross-validation tells the same two-sided story: the model captures the t_{180} ordering usefully (mean absolute percentage error 2.6%, $r = 0.70$) but has no predictive skill yet on the rebound energy ($r = -0.12$), which is consistent with the adversarial review's finding that the rebound channel's noise was understated. We report these as the expected behavior of a five-variable space observed at seven mapped articles rather than as evidence for or against variable relevance; the diagnostics are refreshed every round.

4.3 Later-Round Placeholders. Round-2 outcomes, the budget-matched baseline comparison, the SAASBO-versus-standard-GP calibration audit, and the replicate-article repeatability study are pending data and will complete this section.

5 Discussion

What the seed round establishes. The round-1 data settle three questions the planned-methods stage could only pose. The end-to-end loop closes: printed articles, hundred-drop characterizations, a fitted fully Bayesian surrogate, and a model-recommended batch on the printer within days of the last seed measurement. The objectives discriminate: a 0.169 design-driven range in t_{180} against sub-percent repeatability is a usable signal. And the physics is more adversarial than the cushioning intuition suggests: at this scale and rate, five of the eight measured prisms *amplify* the peak shock reaching their top vertex and only one clearly attenuates it, which sharpens the campaign question from "how much does the architecture cushion" to "which corner of the design space crosses into genuine attenuation, and at what rebound cost."

Honest accounting of the objective stack. We pre-registered an adversarial, data-first review of the objective choices with an external AI referee that was required to commit to its own objective set from the raw campaign files before reading ours. Its findings temper the round-1 formulation and set the round-3 revision: the rebound fraction e_{reb} rests on a ballistic interpretation of the second contact event that has not yet been validated against synchronized video (a restrained-versus-unrestrained check is scheduled); the apparent t_{180}-versus-rebound trade-off is driven substantially by the single strongest attenuator (Spearman $\rho = -0.39$, $p = 0.38$ across the reliable articles); and per-drop standard errors understate article-level uncertainty by roughly an order of magnitude relative to the 0.72% between-article CV measured in a five-print repeatability study. The round-3 plan therefore refits with article-level noise, treats rebound as a candidate constraint pending its validation, and reserves print capacity for replicate articles. We report this trajectory rather than presenting the round-1 choices as final because the loop is the contribution and revising objectives mid-campaign, with the revision criteria stated in advance, is how such loops behave in practice.

Mass as a confound and as a design variable. Holding solid CAD mass constant did not hold printed grams constant: PLA prints at partial infill while thin TPU cables print near solid, so the seed articles span 18.50 g to 22.29 g and printed mass correlates with t_{180} at $r = 0.83$. Because light articles *are* the thick-strut, thin-cable corner by construction, regressing mass out at $n = 8$ would delete the very signal the campaign seeks; the mass-aware rebound objective and the tracked mass metric are the round-1 compromise, and the calibrated printed-mass model (Supplementary Information) enables a constant-printed-mass projection for later rounds.

Relation to prior art. Relative to Pajunen et al. [11], the present work varies the geometry under an optimization loop rather than characterizing a single fabrication condition; relative to Mo et al. [17], the loop's ground truth is physical measurement with simulation demoted to a screening ladder whose predictive reach we quantify rather than assume (Section 3.5). The measured near-unity transmission of most seed articles also cautions against reading the printed-tensegrity literature's attenuation results as generic: the as-printed cells here are not pretensioned, and activating genuine pretension in additively manufactured tensegrity (for example through shrink-on-treatment routes) is an open direction we deliberately scope out of the present campaign.

Practical considerations and path forward. TPU-PLA interface durability under repeated impact (hundred-drop sessions produced no joint failures, an encouraging if unquantified signal), sensitivity to print-process covariates (the print key logs defects, humidity, and the mid-batch nozzle change for exactly this purpose), and transfer to other material pairs remain open. The path forward includes the quasi-static crush characterization (Section 3.3), pretension activation, tiled multi-cell absorbers and lattices [33], and miniaturization to crutch-tip envelopes: long-term crutch use is associ-

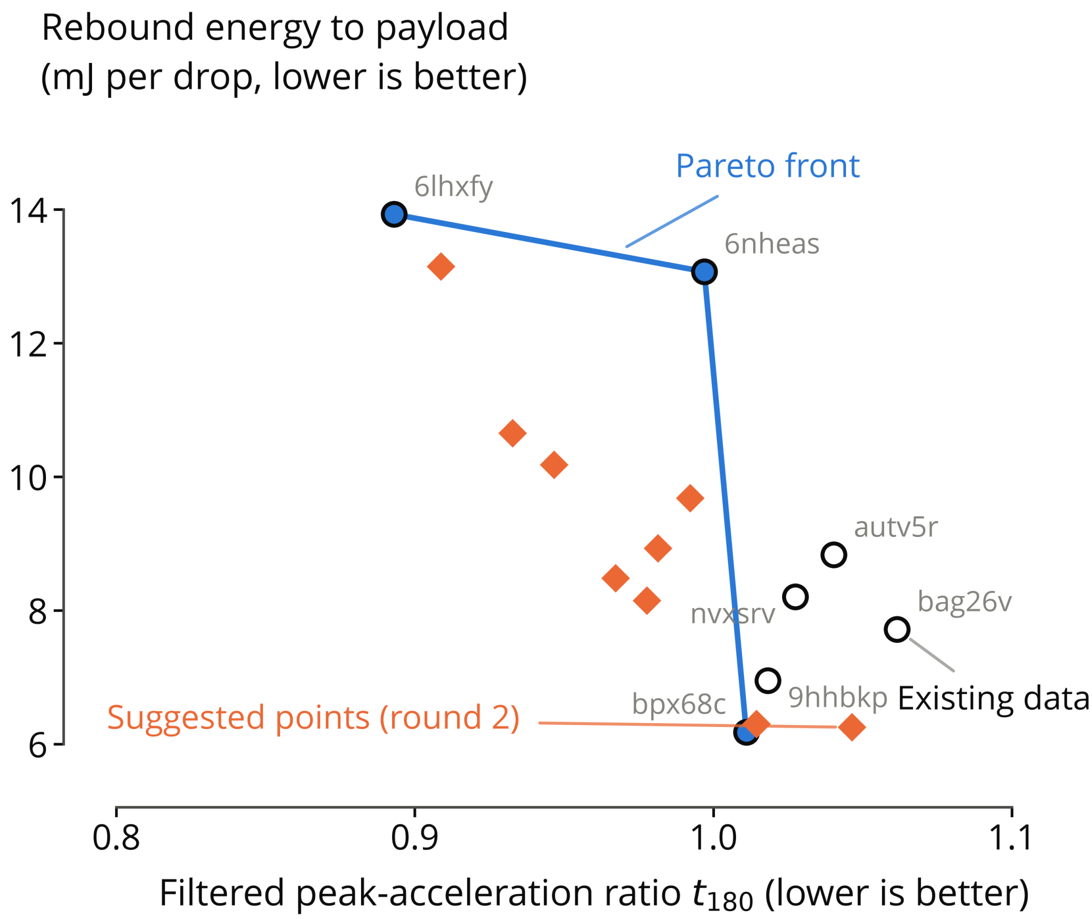

Fig. 7 Measured round-1 objective space. Open circles are tested articles (labeled by print ID) at their stabilized means; the blue markers and line are the observed Pareto front; orange diamonds are the nine model-recommended round-2 designs at their posterior-mean predictions. Both axes improve downward. The observed front and the model-predicted front select the same three articles.

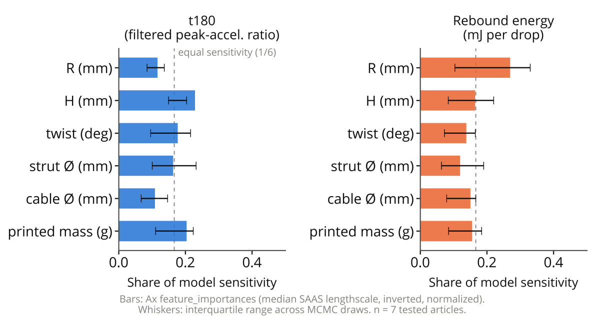

Fig. 8 Round-1 SAASBO feature importances per objective (normalized inverse length scales; median over MCMC draws, interquartile whiskers; inputs are the five design variables plus printed mass). No input separates decisively from the equal-share line (dashed) at seven mapped articles.

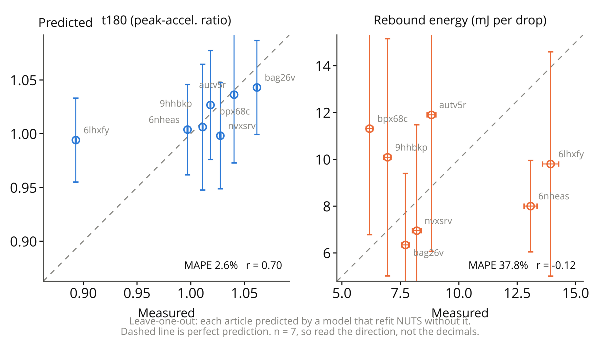

Fig. 9 Round-1 leave-one-out cross-validation of the SAASBO surrogate: predicted versus observed values with model error bars for both objectives.

ated with substantial upper-extremity overuse injury and ground-reaction-force impulses [62–66]; existing spring-loaded [67, 68] and polymer-damped [69] tip designs reduce loading rate but not peak transmitted force, and recent tensegrity crutch concepts [70] and open-source 3D-printed crutch work [71] suggest a plausible translation path, with adjacent footwear and orthotic lattice studies [72–74] providing transferable design heuristics.

Limitations. The evidence base is a single seed round on single prints of each design ($n = 1$ article per geometry, with one triplicate); the drop protocol probes one height, one impact surface, and axial loading only; the specimens never leave the elastic regime, so crush-energy metrics await the quasi-static campaign; per-specimen energy absorption in joules is not recoverable from the current rig (the falling-mass and displacement channels required for a force integral are not instrumented); one tested article lacks a design mapping and one seed design's official article assignment carries a recorded discrepancy; and all results are from a single printer, material pair, and cell topology. The metal-analog protocol (Section 3.6) and the pre-registered revision plan above are the designed responses to these limits.

6 Conclusions

We have presented an experiment-driven Bayesian-optimization framework for designing multi-material 3D-printed tensegrity-inspired energy absorbers using PLA struts and TPU tension elements, and we have closed its first loop on physical hardware. The framework parameterizes a five-variable T_3-prism family under a constant-mass fairness projection, fabricates each candidate in a single multi-material FDM build with TPU tendons anchored inside the PLA strut ends, characterizes every article with 101 instrumented drops, and updates a fully Bayesian Gaussian-process surrogate [22, 24] that recommends the next batch. In the nine-design seed round the filtered peak-acceleration ratio spanned 0.89 to 1.06 against sub-percent repeatability, one design crossed into genuine attenuation, and the first model-recommended batch of nine designs has been generated and sent to fabrication. The approach is broadly applicable to architected absorbers whose constitutive response is difficult to calibrate from first principles and where physical experiments, rather than simulations, are the natural ground truth.

Acknowledgment

This work was supported by the Brigham Young University Mentored Research Grant program. The authors thank the undergraduate researchers who carried out the multi-material printing and impact testing, and the graduate co-mentor who supported day-to-day operations.

Funding Data

- Brigham Young University Mentored Research Grant.

Conflict of Interest

The authors declare no conflict of interest.

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